EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

This introductory lecture establishes the physical, mathematical, and architectural foundations of Digital Signal Processing (DSP). In undergraduate Electrical and Electronics Engineering (EEE), students transition from continuous-time physical systems (analog voltages, currents, electromagnetic waves) to sampled, quantized, discrete-time representations processed by digital microprocessors, DSP chips, and FPGAs.

Pedagogical Strategy:

1. Contrast continuous-time physical signals $x_a(t)$ with discrete-time sequences $x[n]$. Emphasize that n is an integer index, not physical time.
2. Formulate the fundamental sampling relationship $t = nT_s = n/F_s$.
3. Classify discrete-time signals mathematically and physically: Energy signals vs. Power signals, Periodic vs. Aperiodic sequences, and Even vs. Odd symmetric components.
4. Define and visualize elementary discrete sequences: Unit impulse $\delta[n]$, Unit step $u[n]$, Unit ramp $r[n]$, Real exponential $a^n u[n]$, and Complex exponential $e^{j\omega_0 n}$.
5. Highlight the fundamental distinction between continuous and discrete sinusoids: Discrete sinusoidal frequencies are unique only over an interval of length 2π (e.g., $[-\pi, \pi]$ or $[0, 2\pi]$).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Differentiate** between continuous-time, discrete-time, and digital signals across time-domain and amplitude-domain representations.
2. **Compute** the fundamental period N_0 of discrete-time sinusoids and verify periodicity conditions.
3. **Decompose** an arbitrary discrete-time sequence into its even symmetric $x_e[n]$ and odd symmetric $x_o[n]$ components.
4. **Evaluate** the total signal energy E and average power P of elementary and composite discrete-time sequences.
5. **Formulate** elementary signal operations including time-shifting, time-reversal, amplitude scaling, and downsampling/upsampling.

2. MATHEMATICAL FOUNDATIONS & DEFINITIONS

2.1 Continuous to Discrete Sampling

A discrete-time signal $x[n]$ is obtained by sampling a continuous-time signal $x_a(t)$ at uniform intervals of T_s seconds:

$$x[n] = x_a(nT_s) = x_a(t)|_{t=nT_s}, \quad n \in \mathbb{Z}$$

Where:

- $T_s = 1/F_s$ is the sampling period in seconds.
- F_s is the sampling frequency in Hertz (Hz or samples/sec).
- Ω denotes continuous-time angular frequency ($\mathrm{rad/s}$), and $\omega = \Omega T_s = \Omega / F_s$ denotes normalized discrete-time angular frequency ($\mathrm{rad/sample}$).

2.2 Elementary Discrete-Time Sequences

1. **Unit Impulse (Kronecker Delta):**
   $$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$
   Sifting Property: $\sum_{n=-\infty}^{\infty} x[n]\delta[n-n_0] = x[n_0]$.
2. **Unit Step Sequence:**
   $$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = \sum_{k=0}^{\infty} \delta[n-k] = \sum_{m=-\infty}^{n} \delta[m]$$
   Relation to impulse: $\delta[n] = u[n] - u[n-1]$.
3. **Unit Ramp Sequence:**
   $$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$
4. **Real Exponential Sequence:**
   $$x[n] = a^n u[n] = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$
   - For $0 < a < 1$: decaying monotonic sequence.
   - For $-1 < a < 0$: alternating decaying sequence.
   - For $|a| > 1$: exponentially growing sequence (unstable).
5. **Complex Exponential and Sinusoidal Sequences:**
   $$x[n] = Ae^{j(\omega_0 n + \phi)} = A\cos(\omega_0 n + \phi) + jA\sin(\omega_0 n + \phi)$$

2.3 Periodicity Condition in Discrete Time

A discrete-time sequence $x[n]$ is periodic with fundamental period $N \in \mathbb{Z}^+$ if and only if $x[n+N] = x[n]$ for all n. For a sinusoid $x[n] = \cos(\omega_0 n)$:

$$\cos(\omega_0(n+N)) = \cos(\omega_0 n + \omega_0 N) = \cos(\omega_0 n) \iff \omega_0 N = 2\pi k, \quad k \in \mathbb{Z}^+$$
$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q} \quad (\text{Ratio of frequency to } 2\pi \text{ must be a rational number})$$

The fundamental period is $N_0 = \frac{2\pi k}{\omega_0}$, where k is the smallest positive integer such that N_0 is an integer.

2.4 Energy and Power Classification

- **Total Signal Energy E:**

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- **Average Signal Power P:**

$$P = \lim_{K\to\infty} \frac{1}{2K+1} \sum_{n=-K}^{K} |x[n]|^2$$

For a periodic sequence with fundamental period N_0:

$$P = \frac{1}{N_0} \sum_{n=0}^{N_0-1} |x[n]|^2$$

- **Classification:**
    1. *Energy Signal:* $0 < E < \infty \implies P = 0$.
    2. *Power Signal:* $0 < P < \infty \implies E = \infty$.
    3. *Neither:* $E = \infty$ and $P = \infty$ or $P = 0$.

2.5 Even and Odd Symmetry Decomposition

Any arbitrary sequence $x[n]$ can be uniquely decomposed into even and odd components:

$$x[n] = x_e[n] + x_o[n]$$

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]) \quad (\text{Even Component: } x_e[-n] = x_e[n])$$

$$x_o[n] = \frac{1}{2}(x[n] - x[-n]) \quad (\text{Odd Component: } x_o[-n] = -x_o[n], ; x_o[0] = 0)$$

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3. WORKED NUMERICAL EXAMPLES

Example 1.1: Periodicity Analysis of Composite Discrete Sinusoids

Problem: Determine if the following signals are periodic. If periodic, find the fundamental period N_0: (a) $x_1[n] = \cos\left(\frac{3\pi}{7}n + \frac{\pi}{4}\right)$ (b) $x_2[n] = \cos(0.8n)$ (c) $x_3[n] = \cos\left(\frac{\pi}{6}n\right) + \sin\left(\frac{\pi}{8}n\right)$

Solution: (a) $\omega_1 = \frac{3\pi}{7}$.

$$\frac{\omega_1}{2\pi} = \frac{3\pi/7}{2\pi} = \frac{3}{14} \in \mathbb{Q}$$

Since the ratio is rational, $x_1[n]$ is periodic. Fundamental period: $N_1 = \frac{2\pi \cdot k}{\omega_1} = \frac{2\pi \cdot 3}{3\pi/7} = 14$ samples (with $k = 3$).

(b) $\omega_2 = 0.8 = \frac{4}{5}$.

$$\frac{\omega_2}{2\pi} = \frac{0.8}{2\pi} = \frac{0.4}{\pi} \notin \mathbb{Q}$$

Because π is irrational, $\frac{\omega_2}{2\pi}$ is irrational. Therefore, $x_2[n]$ is **aperiodic (non-periodic)**.

(c) For composite signal $x_3[n]$:

- For $\cos\left(\frac{\pi}{6}n\right)$: $\omega_a = \pi/6 \implies \frac{\omega_a}{2\pi} = \frac{1}{12} \implies N_a = 12$.
- For $\sin\left(\frac{\pi}{8}n\right)$: $\omega_b = \pi/8 \implies \frac{\omega_b}{2\pi} = \frac{1}{16} \implies N_b = 16$. Both terms are periodic. The overall fundamental period is the Least Common Multiple (LCM):

$$N_0 = \text{LCM}(N_a, N_b) = \text{LCM}(12, 16) = 48 \text{ samples}$$

Example 1.2: Energy and Power Calculation

Problem: Classify the following signals as Energy, Power, or Neither, and compute the corresponding E or P: (a) $x[n] = (0.6)^n u[n]$ (b) $x[n] = e^{j(\frac{\pi}{4}n + \frac{\pi}{3})}$

Solution: **(a)** Total Energy:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=0}^{\infty} |(0.6)^n|^2 = \sum_{n=0}^{\infty} (0.36)^n$$

Using the infinite geometric series formula $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for $|r| < 1$:

$$E = \frac{1}{1-0.36} = \frac{1}{0.64} = \frac{25}{16} = 1.5625 \text{ Joules}$$

Since $0 < E < \infty$, $x[n]$ is an **Energy Signal** with average power $P = 0$.

(b) The signal has $|x[n]| = \left|e^{j(\frac{\pi}{4}n + \frac{\pi}{3})}\right| = 1$ for all n. Fundamental period: $\omega_0 = \pi/4 \implies N_0 = \frac{2\pi}{\pi/4} = 8$. Average Power:

$$P = \frac{1}{N_0} \sum_{n=0}^{N_0 - 1} |x[n]|^2 = \frac{1}{8} \sum_{n=0}^{7} 1^2 = \frac{8}{8} = 1 \text{ Watt}$$

Total energy $E = \sum_{n=-\infty}^{\infty} 1 = \infty$. Therefore, $x[n]$ is a **Power Signal** with $P = 1$ W and $E = \infty$.

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Define elementary discrete-time sequences: $\delta[n]$, $u[n]$, and $r[n]$. Express $u[n]$ in terms of $\delta[n]$ and vice-versa. *(5 Marks)* **(b)** A continuous-time signal $x_a(t) = 3\cos(100\pi t) + 2\sin(250\pi t)$ is sampled at $F_s = 200$ Hz.

1. Find the discrete-time sequence $x[n]$. *(3 Marks)*
2. Determine whether $x[n]$ is periodic. If periodic, find the fundamental period N_0. *(4 Marks)*
3. Calculate the average power P of $x[n]$. *(3 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Definitions of $\delta[n], u[n], r[n]$ with equations and sketches *(3 Marks)*
  - $u[n] = \sum_{k=-\infty}^{n} \delta[k] = \sum_{m=0}^{\infty} \delta[n-m]$ and $\delta[n] = u[n] - u[n-1]$ *(2 Marks)*
- **Part (b.1):**
  - Substitute $t = n/F_s = n/200$:

    $$x[n] = 3\cos\left(100\pi \frac{n}{200}\right) + 2\sin\left(250\pi \frac{n}{200}\right) = 3\cos\left(\frac{\pi}{2}n\right) + 2\sin\left(\frac{5\pi}{4}n\right)$$

    Since $\frac{5\pi}{4} = 2\pi - \frac{3\pi}{4}$, the principal alias is:

    $$x[n] = 3\cos(0.5\pi n) - 2\sin(0.75\pi n)$$

    (3 Marks)
- **Part (b.2):**
  - $\omega_1 = 0.5\pi \implies \frac{\omega_1}{2\pi} = \frac{1}{4} \implies N_1 = 4.$

- $\omega_2 = 0.75\pi \implies \frac{\omega_2}{2\pi} = \frac{3}{8} \implies N_2 = 8.$
- $N_0 = \mathrm{LCM}(4, 8) = 8$ samples. *(4 Marks)*

- **Part (b.3):**
  - Average power:

$$P = \frac{3^2}{2} + \frac{2^2}{2} = 4.5 + 2.0 = 6.5\ \mathrm{Watts}$$

(3 Marks)

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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Sampling verification
Fs = 200
n = np.arange(0, 16)
xn = 3 * np.cos(np.pi / 2 * n) + 2 * np.sin(5 * np.pi / 4 * n)

# Compute fundamental period power
N0 = 8
P = np.mean(xn[:N0]**2)
print(f"Discrete Signal Power P = {P:.4f} W (Theoretical = 6.5 W)")
```